THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** $(2,\ 2)$	$\left(\frac{-3+(7)}{2},\ \frac{5+(-1)}{2}\right)$
2	**(d)** 1	highest power of the variable is 1
3	**(a)** rational	ratio of integers
4	**(a)** $(x-9)(x+4)$	find a, b with $a+b$ = middle coefficient and ab = constant
5	**(b)** 16 cm	half-chord $= \sqrt{10^2-6^2} = 8$, so chord $= 2\times 8 = 16$
6	**(a)** 33 cm	$\frac{90}{360}\times 2\pi r = 33$
7	**(d)** $\frac{3}{10}$	6 favourable out of 20 equally likely balls
8	**(c)** 96	$ar^{n-1} = 3\times(2)^5 = 96$
9	**(c)** $n = 37$	$7n+2 = 261 \Rightarrow n = 37$
10	**(c)** 4 factors	pq has $1, p, q, pq$
11	**(d)** If the diagonals of a quadrilateral bisect each other, then it is a rhombus.	swap the 'if' part and the 'then' part
12	**(d)** 62	$\frac{50\times 5+80\times 3+65\times 2}{10} = 62$
13	**(b)** 30%	cooling share $= \frac{300}{1000}\times 100 = 30\%$
14	**(c)** 15	groups of equal size, so add the averages and divide by their number
15	**(d)** 53999	values 2, 3, …, min(m, n)
16	**(a)** 10	the chain of reductions ends at gcd(2, 0)
17	**(d)** 20 cm	AG : GD = 2 : 1, so AG = 2/3 of AD
18	**(c)** 3/2	rise ÷ run
19	**(c)** -3	put x = 0
20	**(d)** 264 cm^2	$2\pi rh = 264$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$\frac{-4+x}{2} = 2 \Rightarrow x = 8$; $\frac{3+y}{2} = -1 \Rightarrow y = -5$. So $B = (8,\ -5)$	1+1
22	Average $= \frac{5}{8}$; average of $\frac{1}{2}$ and $\frac{5}{8} = \frac{9}{16}$. So $\frac{9}{16}$ and $\frac{5}{8}$ (any two valid).	1+1
23	$(x-3)(x-2)$	2
24	The hexagon's perimeter is $6r = 3\times(2r) = 3D$, and the circle is longer than the polygon inside it, so $C > 3D$, i.e. $\pi > 3$.	1+1
25	It tells how important or how big each value is relative to the others (1), so each value is multiplied by its weight before dividing by the total weight (1).	1+1
26	Pantograph (copies drawings to scale) (1); Watt's parallel-motion linkage of the steam engine (1).	1+1
27	Multiplying $x+2y=4$ by 3 gives $3x+6y=12$ (1). So the two equations are multiples of each other and every solution of one is a solution of the other (1).	1+1
28	$V = \frac{1}{3}\times$ area of base $\times$ height $= \frac{1}{3}\times 25\times 12$ (1) $= 100$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Radius $= \sqrt{1+64} = \sqrt{65}$ (check $16+49=65$, $49+16=65$ ✓). $OD^2 = 25+36 = 61 < 65 \Rightarrow (-5, 6)$ lies **inside** the circle.	1+1+1
30	Multiply numerator and denominator by 5^2 to get 10^3 in the denominator ⇒ exactly 3 decimal places (the last digit is non-zero).	1+2
31	$x^2+\frac{1}{x^2} = \left(x+\frac{1}{x}\right)^2 - 2 = 9-2 = 7$. $\left(x-\frac{1}{x}\right)^2 = x^2+\frac{1}{x^2}-2 = 5$.	1+1+1

32	$\angle A + \angle D = 180^\circ$ (co-interior) and $\angle A + \angle C = 180^\circ$ (cyclic) ⇒ $\angle C = \angle D$ (2). Base angles equal ⇒ $AD = BC$ (1).	2+1
33	(i) A rectangle (the diagonals are perpendicular) (1). (ii) EF $= GH = \frac{1}{2}AC = 12$ cm; FG $= HE = \frac{1}{2}BD = 5$ cm (1). (iii) $2(12 + 5) = 34$ cm (1).	1+1+1
34	Area $= 144$ cm^2 (1). Side $= \sqrt{144} = 12$ cm (1). (Construction: the mean proportional of 16 and 9.)	2+1
35	For few trials the relative frequency can differ much from the theoretical value, but as the number of trials grows it tends to get closer to the theoretical probability (e.g. tosses of a fair coin approach 0.5).	3
	OR: (i) $\frac{2}{52} = \frac{1}{26}$ (ii) $\frac{1}{52}$ (iii) $\frac{36}{52} = \frac{9}{13}$.	1+1+1
36	Converse: If a number is divisible by 3, then the sum of its digits is a multiple of 3 (1½). It is true: the divisibility rule works in both directions, e.g. 741 and $7 + 4 + 1 = 12$ (1½).	1½+1½
	OR: Counterexample: a 6 cm × 4 cm right triangle and a 12 cm × 2 cm right triangle both have area 12 cm² but are not congruent (1½). Converse: If two triangles are congruent, they have equal area — true (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Proposition: equal sides ⇒ equal opposite angles (isosceles triangle theorem); converse: equal angles ⇒ equal opposite sides, proved by the altitude and AAS (2½). (ii) Proposition: "If a number is a multiple of 3, it is a multiple of 5" is false (e.g. 6); its converse "If a number is a multiple of 5, it is a multiple of 3" is also false (e.g. 10) (2½).	2½+2½
	OR: (i) 18=3×6, 24=3×8, 30=3×10 ✓ (1). (ii) If a number is divisible by 3, then it is divisible by 6 (1). (iii) 9, 15, 21 are divisible by 3 but not by 6: counterexamples; 12 is divisible by both (2). (iv) Of 9, 15, 21, 12 only 12 is divisible by 6: $\frac{1}{4}$ (1).	1+1+2+1
38	(i) $\frac{2\times3+5\times4+3\times3}{3+4+3} = \frac{6+20+9}{10} = 3.5$ m (3). (ii) Simple average $= \frac{2+5+3}{3} = 3.33$ m; it ignores the different lengths for which each depth applies (2).	3+2
	OR: (i) $\frac{12\times31+6\times30+2\times31}{31+30+31}$ (2). (ii) $\frac{372+180+62}{92} = \frac{614}{92} \approx 6.67$ mm per day (3).	2+3
39	Multiplying out and cancelling cross terms gives $x^3 + y^3 + z^3 - 3xyz$ (3). Since $1 + 2 - 3 = 0$, the sum $= 3(1)(2)(-3) = -18$ (2). (Check: $1 + 8 - 27 = -18$ ✓.)	3+2
	OR: (i) $(a + b)(a^2 + 2ab + b^2) = a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3 = a^3 + 3a^2b + 3ab^2 + b^3$. (ii) Replace b by $-b$: $a^3 - 3a^2b + 3ab^2 - b^3$. $97^3 = (100 - 3)^3 = 1000000 - 90000 + 2700 - 27 = 912673$.	3+2
40	(i) $A = 500 + 150n$. (ii) ₹1,400. (iii) $500 + 150n = 2000 \Rightarrow n = 10$. (iv) A straight line through $(0, 500)$ and $(10, 2000)$; slope 150 = monthly addition.	1+1+1+2
	OR: (i) $20 + 5x = 70 \Rightarrow x = 10$. (ii) $C(n) = 20 + 10n$. (iii) ₹140. (iv) $20 + 10n = 170 \Rightarrow n = 15$.	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.